$$\int_{-10}^{10}\left(x^5-6{,}2x^3+3{,}5x^2-7x\right)\cos 2x\,dx=$$

$$=\sin 2x\left(\frac{x^5-6{,}2x^3+3{,}5x^2-7x}{2}-\frac{20x^3-37{,}2x+7}{8}+\frac{120x}{32}\right)+$$

$$+\cos 2x\left(\frac{5x^4-18{,}6x^2+7x-7}{4}-\frac{60x^2-37{,}2}{16}+\frac{120}{64}\right)\Bigg|_{-10}^{10}=$$

$$=\sin 2x\left(0{,}5x^5-5{,}6x^3+1{,}75x^2+4{,}9x-0{,}875\right)+$$

$$+\cos 2x\left(1{,}25x^4-8{,}4x^2+1{,}75x+2{,}45\right)\Big|_{-10}^{10}=$$

$$=2\sin 2x\left(1{,}75x^2-0{,}875\right)+2\cos 2x\left(1{,}75x\right)\Big|_{-10}^{10}=$$

$$=2\sin 20\left(175-0{,}875\right)+2\cos 20\left(17{,}5\right)=$$

[illegible]

$$=348{,}25\sin(20)+35\cos(20)\approx\boxed{332{,}22}$$